Code Explanation

Explanation:

1. Load Patient Data:

- The `load_patients_for_summary_data()` function checks if the `hospital.json` file exists and loads the data from it. If the data is a list, it returns the list as is. If the data is a dictionary of dictionaries, it converts the values to a list of dictionaries.
- The `load_patients_for_crud_operations()` function also checks if the `hospital.json` file exists and loads the data from it. If the data is a list, it converts it to a dictionary keyed by `patient_id`. If the data is already a dictionary, it returns the dictionary as is.
- The `save_patients_for_crud_operations(data)` function converts the dictionary values to a list and saves the data to the `hospital.json` file. If Firebase Firestore is configured, it also saves the data to Firestore.

2. Prepare Patient Summary:

- The `PreparePatientSummary` subgraph loads the patient data for summary, initializes empty lists for patient records, doctors, and diagnoses, and then iterates through the patient records.
- For each patient, it extracts the patient ID, name, and visits, counts the number of visits, collects the unique doctors seen by the patient, and collects all diagnoses from the patient's visits.
- The patient summary information is stored, and a DataFrame is created from the patient records.

3. Calculate New Features:

- The `CalculateNewFeatures` subgraph performs the following calculations:
- Counts the number of visits for each doctor.
- Finds the most visited doctor and the number of visits.
- Calculates the average visits per doctor.
- Counts the occurrences of each diagnosis.
- Finds the most common diagnosis and its count.

4. Summary Menu:

- The `SummaryMenu` subgraph represents the patient data summary menu, which allows the user to perform the following actions:
- Display patient visit details.
- List doctors consulted by each patient.
- Display the patient with the most visits.
- Display the most visited doctor.
- Display the average visits per doctor.
- Display the most common diagnosis.
- Exit the summary menu.

5. Database Menu:

- The `DatabaseMenu` subgraph represents the patient database management menu, which allows the user to perform the following actions:
- Display patients.
- Add a new patient.
- Update a patient (add a new visit).
- Delete a patient.
- Go to the Features Menu (call `run_summary_menu()`).
- Exit the database management menu.

The overall flow of the code is as follows:

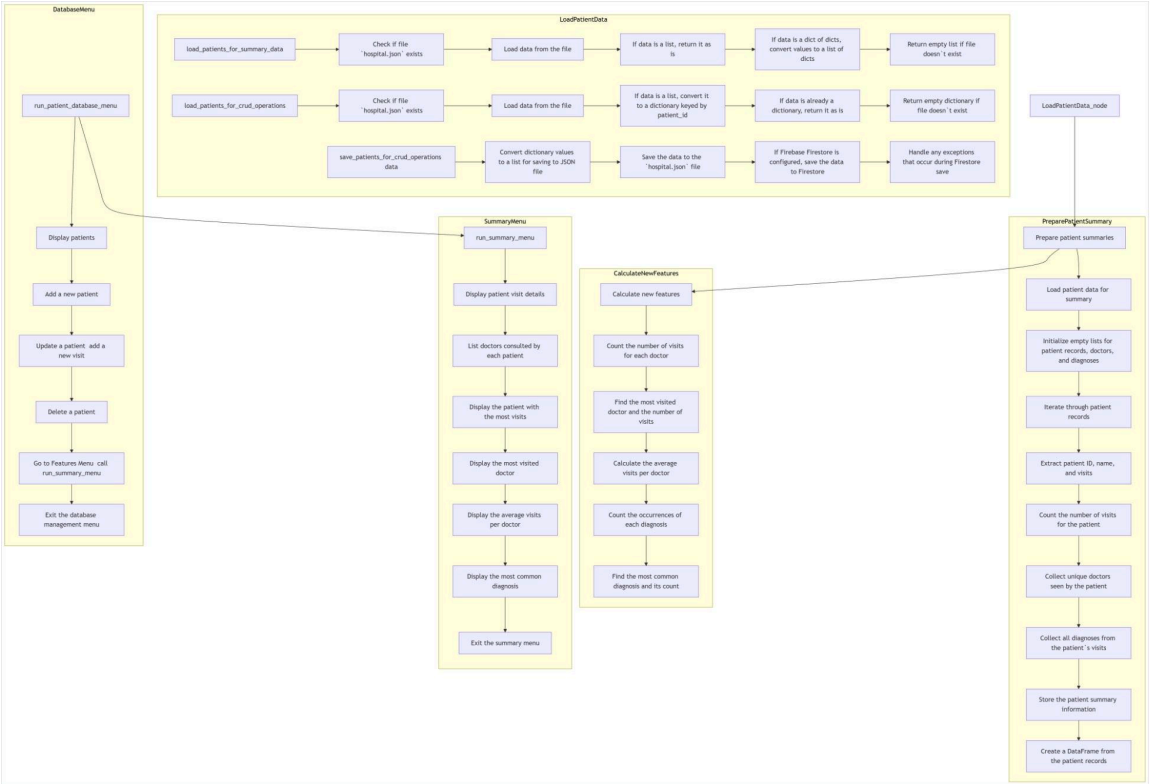
1. The patient data is loaded from the `hospital.json` file for both summary and CRUD operations.

2. The patient summary data is prepared, and new features are calculated based on the patient visit data.

3. The user can access the Summary Menu to view various patient data summaries.

4. The user can also access the Database Menu to perform CRUD operations on the patient data, including saving the updated data to the `hospital.json` file and Firestore (if configured).

5. The Database Menu also provides an option to go to the Features Menu, which calls the `run_summary_menu()` function.


DatabaseMenu
run_patient_database_menu
Display patients
Add a new patient
Update a patient add a new visit
Delete a patient
Go to Features Menu call run_summary_menu
Exit the database management menu
LoadPatientData
load_patients_for_summary_data
Check if file 'hospital.json' exists
Load data from the file
If data is a list, return it as is
If data is a dict of dicts, convert values to a list of dicts
Return empty list if file doesn't exist
load_patients_for_crud_operations
Check if file 'hospital.json' exists
Load data from the file
If data is a list, convert it to a dictionary keyed by patient_id
If data is already a dictionary, return it as is
Return empty dictionary if file doesn't exist
save_patients_for_crud_operations data
Convert dictionary values to a list for saving to JSON file
Save the data to the 'hospital.json' file
If Firebase Firestore is configured, save the data to Firestore
Handle any exceptions that occur during Firestore save
LoadPatientData_node
SummaryMenu
run_summary_menu
Display patient visit details
List doctors consulted by each patient
Display the patient with the most visits
Display the most visited doctor
Display the average visits per doctor
Display the most common diagnosis
Exit the summary menu
CalculateNewFeatures
Calculate new features
Count the number of visits for each doctor
Find the most visited doctor and the number of visits
Calculate the average visits per doctor
Count the occurrences of each diagnosis
Find the most common diagnosis and its count
PreparePatientSummary
Prepare patient summaries
Load patient data for summary
Initialize empty lists for patient records, doctors, and diagnoses
Iterate through patient records
Extract patient ID, name, and visits
Count the number of visits for the patient
Collect unique doctors seen by the patient
Collect all diagnoses from the patient's visits
Store the patient summary information
Create a DataFrame from the patient records